SET 28: VEHICLE PERFORMANCE ANALYSIS

Dataset: Vehicle Performance Data

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Vehicle_ID	Engine_Size (L)	Horsepower	Fuel_Efficiency (km/l)	Top_Speed (km/h)	Safety_Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

PROGRAM 1: Violin Plot of Fuel Efficiency by Safety Rating

Question: Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

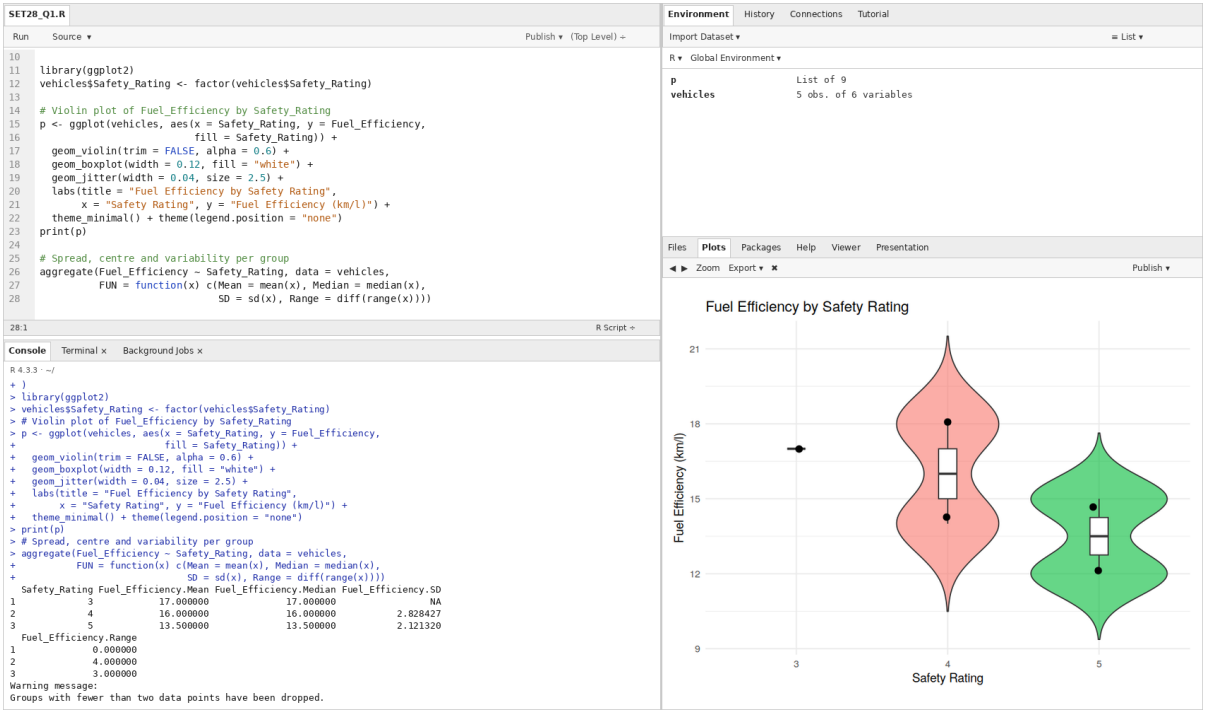
```
# Dataset
vehicles <- data.frame(
  Vehicle_ID      = c("V1", "V2", "V3", "V4", "V5"),
  Engine_Size     = c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower      = c(110, 150, 250, 200, 130),
  Fuel_Efficiency = c(18, 15, 12, 14, 17),
  Top_Speed       = c(180, 200, 250, 220, 190),
  Safety_Rating   = c(4, 5, 5, 4, 3)
)

library(ggplot2)
vehicles$Safety_Rating <- factor(vehicles$Safety_Rating)

# Violin plot of Fuel_Efficiency by Safety_Rating
p <- ggplot(vehicles, aes(x = Safety_Rating, y = Fuel_Efficiency,
                          fill = Safety_Rating)) +
  geom_violin(trim = FALSE, alpha = 0.6) +
  geom_boxplot(width = 0.12, fill = "white") +
  geom_jitter(width = 0.04, size = 2.5) +
  labs(title = "Fuel Efficiency by Safety Rating",
       x = "Safety Rating", y = "Fuel Efficiency (km/l)") +
  theme_minimal() + theme(legend.position = "none")
print(p)

# Spread, centre and variability per group
aggregate(Fuel_Efficiency ~ Safety_Rating, data = vehicles,
          FUN = function(x) c(Mean = mean(x), Median = median(x),
                              SD = sd(x), Range = diff(range(x))))
```

Output 1: Violin Plot of Fuel Efficiency by Safety Rating



Observation: The violin for Safety Rating 4 spans about 14 to 18 km/l (mean and median 16, SD 2.83), and the violin for rating 5 spans about 12 to 15 km/l (mean and median 13.5, SD 2.12). Rating 3 contains only one vehicle (V5, 17 km/l), so ggplot drops its violin (see the warning in the console) and only the point and box line remain. Central tendency falls as the safety rating rises (17, 16, 13.5 km/l), so in this sample the highest-rated vehicles are the least fuel efficient, because the rating 5 cars (V2 and V3) have the larger engines. Rating 4 has the widest spread (range 4 km/l) since it mixes the most efficient car V1 (18) with V4 (14). With only one or two vehicles per group, the violin shapes are indicative rather than reliable density estimates.

PROGRAM 2: Scatter Plot of Horsepower vs Top Speed Coloured by Engine Size

Question: Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.

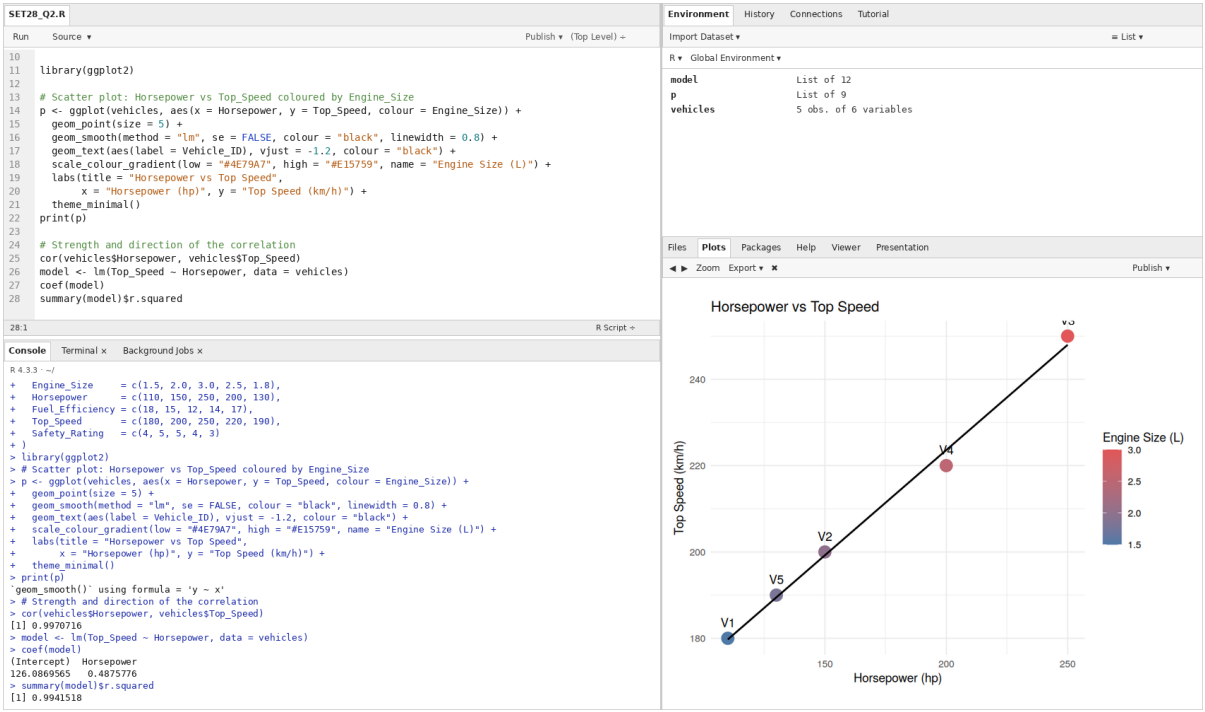
```
# Dataset
vehicles <- data.frame(
  Vehicle_ID      = c("V1", "V2", "V3", "V4", "V5"),
  Engine_Size     = c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower      = c(110, 150, 250, 200, 130),
  Fuel_Efficiency = c(18, 15, 12, 14, 17),
  Top_Speed       = c(180, 200, 250, 220, 190),
  Safety_Rating   = c(4, 5, 5, 4, 3)
)

library(ggplot2)

# Scatter plot: Horsepower vs Top_Speed coloured by Engine_Size
p <- ggplot(vehicles, aes(x = Horsepower, y = Top_Speed, colour = Engine_Size)) +
  geom_point(size = 5) +
  geom_smooth(method = "lm", se = FALSE, colour = "black", linewidth = 0.8) +
  geom_text(aes(label = Vehicle_ID), vjust = -1.2, colour = "black") +
  scale_colour_gradient(low = "#4E79A7", high = "#E15759", name = "Engine Size (L)") +
  labs(title = "Horsepower vs Top Speed",
       x = "Horsepower (hp)", y = "Top Speed (km/h)") +
  theme_minimal()
print(p)

# Strength and direction of the correlation
cor(vehicles$Horsepower, vehicles$Top_Speed)
model <- lm(Top_Speed ~ Horsepower, data = vehicles)
coef(model)
summary(model)$r.squared
```

Output 2: Horsepower vs Top Speed Scatter Plot



Observation: Horsepower and Top Speed have an almost perfect positive linear relationship (r = 0.997, R-squared = 0.994). The fitted line is Top_Speed = 126.09 + 0.4876 x Horsepower, so every extra 10 hp adds about 4.9 km/h. The colour scale shows that engine size rises in step with both variables: the small 1.5 L engine (V1, 110 hp, 180 km/h) is at the lower left in blue, while the 3.0 L engine (V3, 250 hp, 250 km/h) is at the upper right in red. The trend is that larger engines deliver more power and a higher top speed. This is based on five vehicles, so it describes this dataset rather than all vehicles.

PROGRAM 3: Correlation Heatmap of All Numerical Variables

Question: Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.

```
# Dataset
vehicles <- data.frame(
  Vehicle_ID      = c("V1", "V2", "V3", "V4", "V5"),
  Engine_Size     = c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower      = c(110, 150, 250, 200, 130),
  Fuel_Efficiency = c(18, 15, 12, 14, 17),
  Top_Speed       = c(180, 200, 250, 220, 190),
  Safety_Rating   = c(4, 5, 5, 4, 3)
)

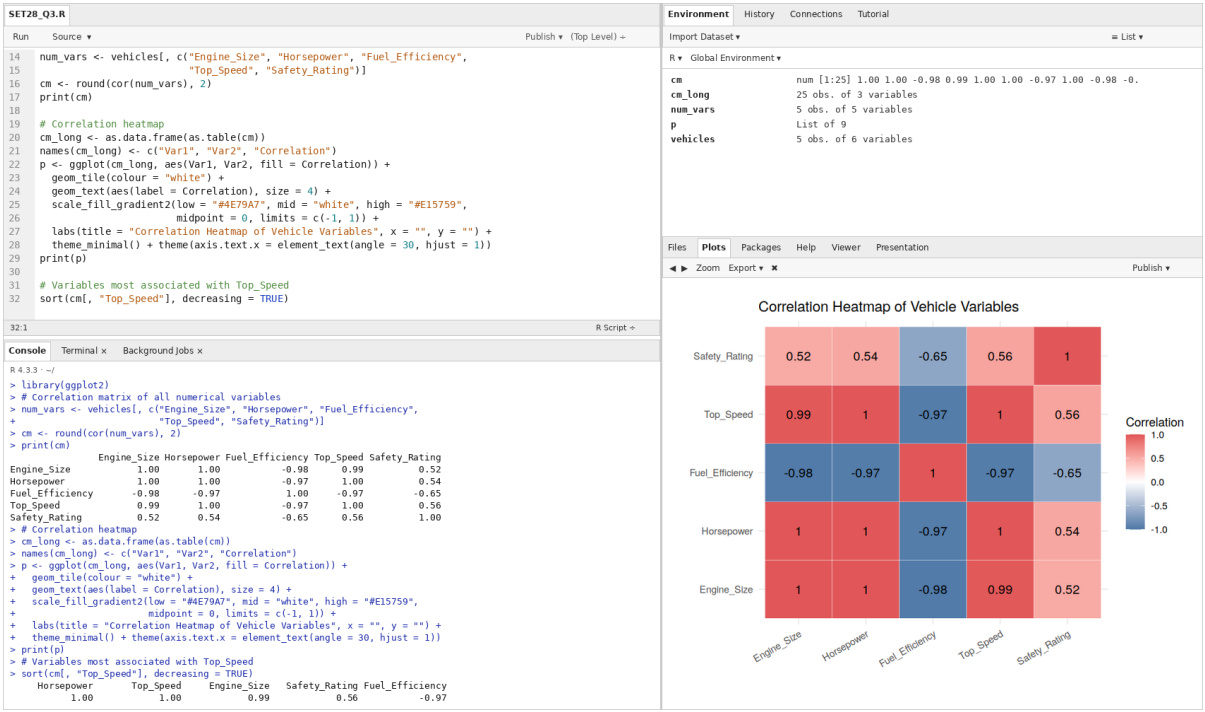
library(ggplot2)

# Correlation matrix of all numerical variables
num_vars <- vehicles[, c("Engine_Size", "Horsepower", "Fuel_Efficiency",
                         "Top_Speed", "Safety_Rating")]
cm <- round(cor(num_vars), 2)
print(cm)

# Correlation heatmap
cm_long <- as.data.frame(as.table(cm))
names(cm_long) <- c("Var1", "Var2", "Correlation")
p <- ggplot(cm_long, aes(Var1, Var2, fill = Correlation)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = Correlation), size = 4) +
  scale_fill_gradient2(low = "#4E79A7", mid = "white", high = "#E15759",
                       midpoint = 0, limits = c(-1, 1)) +
  labs(title = "Correlation Heatmap of Vehicle Variables", x = "", y = "") +
  theme_minimal() + theme(axis.text.x = element_text(angle = 30, hjust = 1))
print(p)

# Variables most associated with Top_Speed
sort(cm[, "Top_Speed"], decreasing = TRUE)
```

Output 3: Correlation Heatmap




```
> library(ggplot2)
> # Correlation matrix of all numerical variables
> num_vars <- vehicles[, c("Engine_Size", "Horsepower", "Fuel_Efficiency",
+                          "Top_Speed", "Safety_Rating")]
> cm <- round(cor(num_vars), 2)
> print(cm)
             Engine_Size Horsepower Fuel_Efficiency Top_Speed Safety_Rating
Engine_Size         1.00       1.00           -0.98      0.99          0.52
Horsepower          1.00       1.00           -0.97      1.00          0.54
Fuel_Efficiency    -0.98      -0.97            1.00     -0.97         -0.65
Top_Speed           0.99       1.00           -0.97      1.00          0.56
Safety_Rating       0.52       0.54           -0.65      0.56          1.00
> # Correlation heatmap
> cm_long <- as.data.frame(as.table(cm))
> names(cm_long) <- c("Var1", "Var2", "Correlation")
> p <- ggplot(cm_long, aes(Var1, Var2, fill = Correlation)) +
+   geom_tile(colour = "white") +
+   geom_text(aes(label = Correlation), size = 4) +
+   scale_fill_gradient2(low = "#4E79A7", mid = "white", high = "#E15759",
+                        midpoint = 0, limits = c(-1, 1)) +
+   labs(title = "Correlation Heatmap of Vehicle Variables", x = "", y = "") +
+   theme_minimal() + theme(axis.text.x = element_text(angle = 30, hjust = 1))
> print(p)
> # Variables most associated with Top_Speed
> sort(cm[, "Top_Speed"], decreasing = TRUE)
  Horsepower    Top_Speed  Engine_Size Safety_Rating Fuel_Efficiency
        1.00         1.00         0.99          0.56          -0.97
```

Observation: Top_Speed is most strongly associated with Horsepower (r = 0.997, displayed as 1.00) and Engine_Size (0.99), both strongly positive. It is strongly negative with Fuel_Efficiency (-0.97) and moderately positive with Safety_Rating (0.56). Engine_Size, Horsepower and Top_Speed are almost perfectly correlated with each other, so they carry nearly the same information about performance. The negative column for Fuel_Efficiency (-0.98 with Engine_Size, -0.97 with Horsepower) shows the central trade-off: bigger, more powerful engines raise speed but lower kilometres per litre. Safety_Rating has only a weak to moderate link with the other variables (0.52 to 0.56, and -0.65 with efficiency), so it is largely independent of raw performance.

PROGRAM 4: Interactive Vehicle Performance Dashboard in Tableau

Question: In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel

efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

Calculated fields (Analysis > Create Calculated Field)

```
// Efficiency Norm (0 to 1, fleet minimum to fleet maximum)
(SUM([Fuel_Efficiency]) - {MIN([Fuel_Efficiency])})
/ ({MAX([Fuel_Efficiency])} - {MIN([Fuel_Efficiency])})

// Speed Norm (0 to 1)
(SUM([Top_Speed]) - {MIN([Top_Speed])})
/ ({MAX([Top_Speed])} - {MIN([Top_Speed])})

// Balance Score (geometric mean: high only when BOTH are high)
SQRT([Efficiency Norm] * [Speed Norm])

// Balance Category
IF [Balance Score] >= 0.35 THEN "Balanced"
ELSEIF [Efficiency Norm] > [Speed Norm] THEN "Efficiency-focused"
ELSE "Performance-focused" END
```

Tableau Steps

1. Open Tableau and connect to the vehicle data file (Excel/CSV). Check that Engine_Size, Horsepower, Fuel_Efficiency and Top_Speed are numeric measures and Vehicle_ID is a dimension.

2. Right-click Safety_Rating and choose Convert to Dimension so it can be used as a category filter.

3. Create the calculated fields Efficiency Norm, Speed Norm, Balance Score and Balance Category (shown above).

4. Create a worksheet named "Balance Score". Drag Vehicle_ID to Columns and Balance Score to Rows, choose Bar, put Balance Category on Color, sort descending and show mark labels.

5. Create a worksheet named "Efficiency by Safety Rating". Drag Safety_Rating to Columns and Fuel_Efficiency to Rows (Average), choose Bar and show mark labels.

6. Create a worksheet named "Efficiency vs Speed". Drag Fuel_Efficiency to Columns and Top_Speed to Rows, put Vehicle_ID on Detail and Label, Horsepower on Size and Balance Category on Color. Choose Circle in the Marks card. Optionally add a Trend Line from the Analytics pane.

7. Add the filters: drag Engine_Size to Filters (Range of values) and Safety_Rating to Filters (select all values). Right-click each filter, choose Show Filter, and set Apply to Worksheets > All Using This Data Source.

8. Click New Dashboard, set the size to Automatic, drag in the three worksheets, and add the title "Vehicle Performance Dashboard".

9. Select the bar chart and enable Use as Filter so clicking a vehicle highlights it in the scatter plot. Show the Engine_Size and Safety_Rating filters and the Balance Category legend.

10. Test the dashboard by moving the Engine_Size slider, unticking Safety_Rating values and hovering over points to read the tooltips.

Balance scoring results

Vehicle_ID	Engine (L)	Fuel Eff. (km/l)	Top Speed (km/h)	Safety	Balance Score	Category
V1	1.5	18	180	4	0.00	Efficiency-focused
V2	2.0	15	200	5	0.38	Balanced
V3	3.0	12	250	5	0.00	Performance-focused
V4	2.5	14	220	4	0.44	Balanced
V5	1.8	17	190	3	0.35	Efficiency-focused

Output 4: Vehicle Performance Dashboard (layout preview)



Observation: V4 (2.5 L, 220 km/h, 14 km/l) and V2 (2.0 L, 200 km/h, 15 km/l) are the vehicles that best balance fuel efficiency and performance, with Balance Scores of 0.44 and 0.38. V2 is the strongest all-round choice because it also has the top Safety Rating of 5. V1 (18 km/l) and V5 (17 km/l) are the most economical but the slowest, and V5 only narrowly misses the Balanced threshold (0.345 against 0.35). V3 has the highest speed (250 km/h) but the lowest efficiency (12 km/l), so its Balance Score is zero. The scatter plot shows the trade-off clearly: moving to the right (more efficient) means moving down (slower), and the Balanced vehicles sit in the middle. The Safety_Rating bar chart shows average efficiency falling from 17 to 16 to 13.5 km/l as rating rises, and the Engine_Size filter lets a buyer narrow the view to a size class. The Balance Score is a simple min-max score calculated across this five-vehicle fleet, so it ranks these vehicles against each other only.

Procedure

1. Create the vehicle data frame (Vehicle_ID, Engine_Size, Horsepower, Fuel_Efficiency, Top_Speed, Safety_Rating) in R.

2. Convert Safety_Rating to a factor and draw the violin plot of Fuel_Efficiency using ggplot2 with geom_violin(), geom_boxplot() and geom_jitter().

3. Draw the scatter plot of Horsepower against Top_Speed coloured by Engine_Size, add a regression line, and compute cor() and lm().

4. Compute the correlation matrix with cor() and draw the heatmap with geom_tile().

5. Import the data into Tableau and create the Balance Score and Balance Category calculated fields.

6. Build the bar chart and scatter plot worksheets and add the Engine_Size and Safety_Rating filters.

7. Combine the worksheets into a dashboard, enable Use as Filter and test the interactivity.

Result

The Vehicle Performance data was successfully analysed in R and Tableau. Fuel efficiency is lower for higher safety-rated vehicles in this sample, Horsepower and Top_Speed are almost perfectly correlated ($r = 0.997$), Top_Speed is most strongly tied to Horsepower and Engine_Size and negatively to Fuel_Efficiency, and the interactive Tableau dashboard identified V4 and V2 as the vehicles that best balance efficiency and performance.